# T3 group factors: EFT_1_after_F_S1_S2

Active theory: SMEFT with the Weinberg operator $C_5$. No T3 heavy field remains dynamical.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

## Notation key

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q}\, Y_u\, \widetilde{H}\, u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q}\, Y_d\, H\, d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L}\, Y_e\, H\, e + \text{h.c.}$ |
| $T$ | $\mathrm{Tr}\left(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d\right)$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $C_5^{ij}$ | $(L_i^T C L_j)\, HH$ |

## Stage field content

| run | $S_1$ | $S_2$ | $F$ | integrated at this stage | active after matching |
|---|---|---|---|---|---|
| T3-B | $(2,-\frac{1}{2})$ | $(2,\frac{1}{2})$ | $(1,0)$ | F, S1, S2 | none |
| T3-C | $(2,-\frac{1}{2})$ | $(2,\frac{1}{2})$ | $(3,0)$ | F, S1, S2 | none |
| T3-A | $(1,0)$ | $(3,1)$ | $(2,\frac{1}{2})$ | F, S1, S2 | none |
| T3-D | $(3,-1)$ | $(1,0)$ | $(2,-\frac{1}{2})$ | F, S1, S2 | none |
| T3-E | $(3,0)$ | $(3,1)$ | $(2,\frac{1}{2})$ | F, S1, S2 | none |

## Decoupling statement

All dependence on the original $S_1$, $S_2$ and $F$ representations is contained in the matched boundary value $C_5(\mu_{\rm th})$. The one-loop running below the final threshold is common to all T3 classes.

$\beta_{C_5}$

| $\mathcal{S}$ | T3 − B ($\alpha=-1$) | T3 − C ($\alpha=-1$) | T3 − A ($\alpha=0$) | T3 − D ($\alpha=-2$) | T3 − E ($\alpha=0$) |
|---|---|---|---|---|---|
| $\lambda_1 C_5$ | 2 | 2 | 2 | 2 | 2 |
| $g_2^2 C_5$ | −3 | −3 | −3 | −3 | −3 |
| $T C_5$ | 2 | 2 | 2 | 2 | 2 |
| $(Y_e^\dagger Y_e)^T C_5$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ |
| $C_5 (Y_e^\dagger Y_e)$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ | $-\frac{3}{2}$ |

## Flavor form

$$16\pi^2 \frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \tag{1}$$

$$- \frac{3}{2}\left[(Y_e^\dagger Y_e)^T C_5 + C_5 (Y_e^\dagger Y_e)\right]. \tag{2}$$